

Tip-of-the-Tongue Retrieval on TREC 2023: A Tuned BERT–GRU Bi-encoder with LoRA Reranking

IS 584 Final Report

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Abstract—Tip-of-the-Tongue (ToT) retrieval is a known-item retrieval task in which a user describes a target item using verbose, often noisy natural-language descriptions with little lexical overlap with the target document. Building on the interim report, this work extends the BERT–GRU bi-encoder proposed there with (i) a 12-configuration tokenizer sweep that varies both subword algorithm and projection dimension (RQ1), (ii) a sentence-dropout sweep over the best tokenizer to study explicit query noise reduction (RQ2), and (iii) a LoRA-tuned Qwen2.5 pointwise reranker that operates on the bi-encoder’s top-100 candidates. On the TREC 2023 ToT development set (150 queries, 231,852 documents), the final pipeline reaches NDCG@1000 of 0.2892 and R@100 of 0.4067. The reranker improves NDCG@1000 over the dense BERT–GRU stage (0.2079) by 39% relative and over the BM25 baseline (0.1165) by 148% relative; because the reranker only reorders the dense top-100, R@100 is unchanged from the dense stage (0.4067) and improves over BM25 (0.1600) by 154% relative. Both a Wilcoxon signed-rank test and a randomised permutation test confirm that the NDCG@1000 gains over both baselines are statistically significant ($p < 10^{-3}$). UMAP analysis and a systematic error analysis on the worst-performing 10% of queries indicate that the remaining gap is dominated by queries whose descriptions are short or contain salient but incorrect plot details. The code can be found here: https://github.com/atg93/project_584

Index Terms—information retrieval, tip-of-the-tongue, dense retrieval, BERT, BM25, LoRA, reranking, interpretability

I. INTRODUCTION

Tip-of-the-Tongue (ToT) retrieval is a known-item retrieval task in which the user knows the target item exists but cannot recall its name. Queries are verbose, multi-sentence natural-language descriptions that may contain false or missing information, with little keyword overlap with the target document [1]. Classical lexical retrieval is ill-suited to this regime, which has motivated the use of dense neural retrievers and, more recently, generative LLM-based rerankers.

This work continues the experimental program defined in the interim report and investigates two research questions on the TREC 2023 ToT track:

- **RQ1:** *Does changing the tokenizer (and its associated encoder) affect retrieval performance on the ToT task?*

- **RQ2:** *Can noise reduction techniques be further improved for retrieval performance?*

The interim report established the BM25 baseline and a first version of the BERT–GRU bi-encoder. This final report contributes: (1) a 12-configuration tokenizer sweep over WordPiece (cased / uncased), BPE, and Unigram tokenizers crossed with projection dimensions $d \in \{256, 512, 768\}$; (2) a sentence-dropout sweep $\in \{0.1, 0.2, 0.3\}$ on the best tokenizer configuration; (3) a LoRA-tuned Qwen2.5 pointwise reranker stacked on top of the best bi-encoder; (4) a UMAP-based interpretability analysis of the dense embedding space; and (5) a systematic error analysis isolating concrete failure modes.

II. DATASET

All experiments use the TREC 2023 Tip-of-the-Tongue track [1]. The collection consists of 231,852 movie documents (Wikipedia plot summaries enriched with title and metadata) and a query set partitioned into training, development, and test splits. Following the interim report, all metric numbers in this report are computed on the official development set (150 queries), since the test labels remain held out.

Queries are markedly longer than in standard ad-hoc IR. A query is composed of N natural-language sentences (mean ≈ 8 , max 30) describing plot fragments, characters, and personal recollections such as “I think it was a horror movie but I’m not sure”. The Reddit TOMT corpus [2] is used as an auxiliary training source providing broader ToT coverage across domains. To prevent contamination of the evaluation distribution, all preprocessing, statistics, and tokenizer training were carried out only on the train + Reddit portions; the dev split was touched only at evaluation time.

III. MODELING

A. BM25 baseline

A standard BM25 implementation indexed over title + plot text serves as the lexical baseline [4]. Hyperparameters k_1 and b were grid-searched over $k_1 \in \{0.5, 1.0, 1.5\}$ and

$b \in \{0.25, 0.5, 0.75, 1.0\}$, yielding 12 configurations; the best-performing ($k_1=1.5, b=0.75$) configuration is the one used in all subsequent comparisons.

B. BERT-GRU bi-encoder

As shown in Figure 1, the proposed bi-encoder uses two asymmetric encoders, reflecting the differing structure of queries and documents. The *query encoder* treats each of the N query sentences independently: every sentence is encoded by BERT [3], and the [CLS] embeddings form a length- N sequence. Because BERT encodes each sentence in isolation, it discards the sequential structure that is characteristic of Tip-of-the-Tongue descriptions, where users recall a film as an ordered narrative (setup, salient details, then ending). To recover this structure, a Gated Recurrent Unit (GRU) [5] consumes the sequence and models the temporal dependencies between sentences, producing hidden states that encode each clue in the context of those preceding it. A sentence dropout layer randomly zeros entire sentence embeddings with probability p during training; this acts as a noise-reduction mechanism by forcing the GRU to form a coherent query representation from incomplete clue sets, suppressing reliance on any single noisy or off-topic sentence and improving robustness to the vague or partial descriptions typical of real users. A learned attention head then pools over the GRU hidden states by scoring each local state against the final hidden state—which serves as a global query summary—allowing the model to dynamically up-weight the most discriminative clues rather than treating all sentences equally, as mean-pooling would. Crucially, the GRU also compresses a variable-length sentence sequence into a single fixed-size representation, which a linear projection maps to a d -dimensional embedding suitable for similarity search. The *document encoder*, by contrast, requires no recurrence: documents are single coherent texts, so a standard BERT [3] [CLS] pooler with a linear projection to the same d -dimensional space suffices. Retrieval is performed with FAISS [6] inner-product search over L2-normalized vectors, equivalent to cosine similarity. Training uses in-batch contrastive loss with temperature $\tau = 0.05$ [7], alternating TREC ToT and Reddit TOMT batches each epoch.

C. Tokenizer sweep (RQ1)

RQ1 is operationalised as a fully crossed sweep over four tokenizer / encoder pairs and three projection dimensions: WORDPIECE-UNCASED (bert-base-uncased), WORDPIECE-CASED (bert-base-cased), BPE (roberta-base), and UNIGRAM (albert-base-v2) crossed with $d \in \{256, 512, 768\}$. All other architectural choices, learning rate schedule, training data, and the contrastive loss are held fixed so that the sweep isolates the tokenizer / encoder contribution. The 12 runs were logged to a single W&B project (tot-tokenizer-sweep).

D. Sentence-dropout sweep (RQ2)

RQ2 is studied with the best tokenizer / dimension from RQ1 fixed (WORDPIECE-UNCASED, $d=512$). Sentence

dropout p is swept over $\{0.1, 0.2, 0.3\}$, leaving everything else unchanged. This isolates how aggressively the model should be regularised against unreliable sentences in ToT descriptions.

E. LoRA-tuned Qwen2.5 reranker

A two-stage cascade is added on top of the best bi-encoder, trading the bi-encoder’s retrieval speed for deeper query–document interaction where it is affordable. The dense retriever returns the top-1000 candidates; the top 100 are then re-scored by a pointwise LoRA-tuned Qwen2.5 reranker. Unlike the bi-encoder, which compares the query and document only through a single dot product over independently encoded vectors, the reranker places the user description and the candidate movie in a *single prompt*, as shown in Figure 2, so every query token can attend to every document token. This full cross-attention captures fine-grained plot- and detail-level matches that a compressed embedding similarity cannot represent, while the pretrained Qwen2.5 backbone contributes broad world knowledge and reasoning over movie content. Applying this expensive cross-encoder only to the 100 most promising candidates keeps the cascade tractable, since scoring the full corpus this way would be prohibitive.

The reranker is a 4-bit quantized Qwen2.5 base model, wrapped with a LoRA adapter ($r=16$, $\alpha=32$, dropout= 0.1) and trained on (query, candidate, label) triples. Because only the LoRA adapters are updated, the base model’s pretrained knowledge is preserved while adaptation remains parameter-efficient, producing small checkpoints and allowing the model to run under 4-bit quantization on modest hardware. The relevance score reuses the model’s own language-modeling head and is computed as $\log P(\text{“yes”}) - \log P(\text{“no”})$ given a prompt that contains the user description and the candidate movie’s title + truncated plot, yielding a calibrated pointwise signal without any separately trained classification head. Training uses a cosine learning-rate schedule with warmup, $\text{lr}_{\max}=5 \times 10^{-5}$, and bf16 mixed precision. The combined score is a linear blend $s = \alpha s_{\text{DENSE}} + (1-\alpha) s_{\text{RERANKER}}$ with $\alpha = 0.3$, tuned on a held-out slice of the training set.

IV. EVALUATION PROTOCOL

Following the official TREC 2023 ToT evaluation [1], NDCG@1000 is reported as the primary metric and R@100 as the secondary metric (which captures the recall pool fed to the reranker). Per-query scores are cached so that pairwise comparisons can be carried out without re-running retrieval. Improvements over a comparator are tested with the Wilcoxon signed-rank test [8] on per-query NDCG@1000. All runs use a fixed seed (`torch.manual_seed(42)`, `np.random.seed(42)`, and FAISS deterministic settings) and are logged in W&B for reproducibility.

V. RESULTS

A. RQ1: Effect of tokenizer and projection dimension

Table I reports the 12 tokenizer / dimension configurations sorted by NDCG@1000. The strongest configuration is WORDPIECE-UNCASED at $d=512$ (NDCG@1000 = 0.1203,

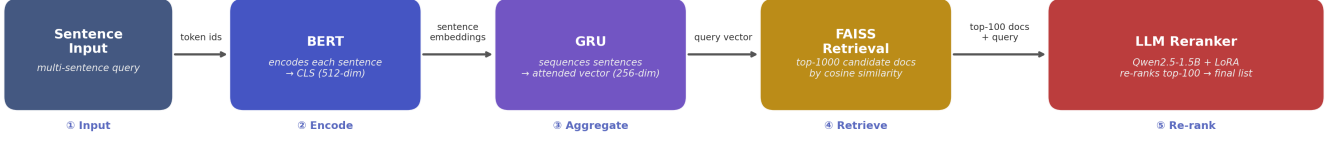


Fig. 1. Current Architecture for Tip-of-the-Tongue Retrieval

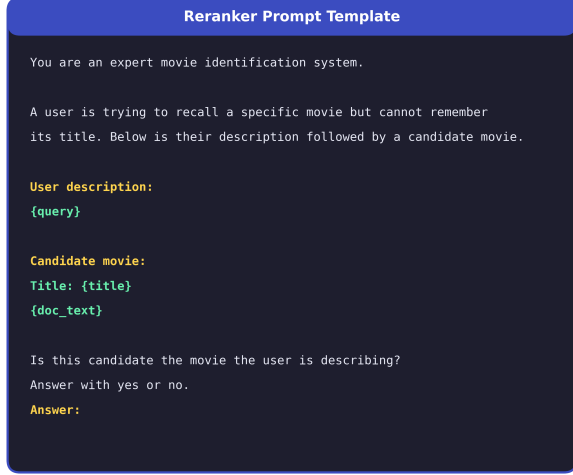


Fig. 2. Prompt Template of the LLM Reranker

$R@100 = 0.2400$), narrowly ahead of WORDPIECE-CASED at the same dimension (0.1181); this configuration is the one carried forward into RQ2. Ranking the encoder families by their best configuration gives WORDPIECE-UNCASED > WORDPIECE-CASED > UNIGRAM > BPE, with the two WordPiece variants clearly separated from UNIGRAM and BPE. For the WordPiece encoders, $d=512$ matches or exceeds $d=768$ (e.g. uncased: 0.1203 vs. 0.1064), suggesting that the linear projection head saturates before the largest dimensionality. Note that these sweep runs isolate the tokenizer/encoder contribution using the BERT encoder alone, without the GRU aggregation layer; this is why the best sweep configuration (0.1203) matches the “BERT-only fine-tuned (wordpiece)” row in Table III. The GRU and the attention pooler are added only to the final configuration, which—after the longer training schedule detailed in Sec. V-B—raises NDCG@1000 to 0.2079 (Table III).

Several BPE and UNIGRAM configurations—including all three BPE dimensions and UNIGRAM at $d=512$ —collapse to a NDCG@1000 of 0.0000. We attribute this to under-training rather than to a genuine property of these tokenizers: each of the 12 configurations was trained for only 3 epochs to keep the full crossed sweep tractable within the project timeline, and the BPE and UNIGRAM encoders appear not to have converged within that budget.

TABLE I

RQ1 — TOKENIZER / DIMENSION SWEEP ON THE TREC 2023 ToT DEVELOPMENT SET, SORTED BY NDCG@1000. BOLD MARKS THE CONFIGURATION CARRIED INTO RQ2.

Tokenizer	d	NDCG@1000	R@100
WordPiece-uncased	768	0.1064	0.2133
WordPiece-cased	768	0.0759	0.1466
BPE	768	0.0000	0.0000
Unigram	768	0.0637	0.1333
WordPiece-uncased	512	0.1203	0.2400
WordPiece-cased	512	0.1181	0.2933
BPE	512	0.0000	0.0000
Unigram	512	0.0000	0.0000
WordPiece-uncased	256	0.0999	0.2800
WordPiece-cased	256	0.1096	0.2400
BPE	256	0.0000	0.0000
Unigram	256	0.0788	0.1933

B. RQ2: Sentence-dropout sensitivity

Holding the encoder fixed at WORDPIECE-UNCASED, $d=512$, and adding the GRU aggregation layer (on which the sentence-dropout regulariser operates), Table II sweeps sentence dropout over $\{0.1, 0.2, 0.3\}$. Dropout $p=0.1$ is the strongest configuration (NDCG@1000 = 0.1543, $R@100 = 0.3400$), outperforming both the interim default $p=0.2$ (0.1140) and the more aggressive $p=0.3$ (0.1187). The ordering indicates that, on this dense backbone, light regularisation is preferable: a small amount of sentence dropout supplies enough robustness to incomplete clue sets without discarding signal, whereas heavier dropout ($p \geq 0.2$) starves the GRU of usable sentences during training and degrades the query representation.

Although $p=0.1$ is the strongest configuration in the dropout sweep, the final pipeline reported in Table III uses $p=0.2$: the $p=0.1$ run had not finished training in time for the submission deadline, so the $p=0.2$ checkpoint—the interim default—was retained for the end-to-end experiments. Note that the sweep runs in Table II are trained for 10 epochs each, so that the three dropout settings can be compared at equal cost, whereas the retained $p=0.2$ checkpoint has accumulated 100 training epochs since the interim phase. This difference in training budget—not a discrepancy—is why the BERT+GRU row in Table III (0.2079) substantially exceeds the corresponding 10-epoch sweep entry in Table II (0.1140). The reranker and ablation results in Table III should therefore be read as built on the fully trained $p=0.2$ bi-encoder, and the small gap to $p=0.1$ is left as an immediate follow-up.

TABLE II
RQ2 — SENTENCE-DROPOUT SWEEP.

Sentence dropout p	NDCG@1000	R@100
0.1	0.1543	0.3400
0.2	0.1140	0.2466
0.3	0.1187	0.2266

TABLE III
ABLATION STUDY ACROSS PIPELINE STAGES.

Model	NDCG@1000	R@100
BM25 ($k_1=1.5, b=0.75$)	0.1165	0.1600
Pretrained BERT (zero-shot)	0.0000	0.0000
BERT-only fine-tuned (wordpiece)	0.1203	0.2400
BERT + GRU	0.2079	0.4067
BERT + GRU + Qwen2.5-0.5B-LoRA rerank	0.2892	0.4067
BERT + GRU + Qwen2.5-1.5B-LoRA rerank	0.2646	0.4067

C. Full pipeline with LoRA reranker

Table III reports the end-to-end pipeline. The Qwen2.5 reranker is restricted to the top-100 candidates of the tuned bi-encoder, so R@100 is bounded by the dense stage; the reranker can only *reorder* within that pool. Both reranker sizes substantially improve NDCG@1000. The smaller Qwen2.5-0.5B variant is in fact the stronger reranker on the development set despite one-third the parameters and roughly half the rerank latency of the 1.5B variant, and is therefore adopted as the final model. R@100 is unchanged because the candidate set is fixed by the dense retriever.

D. Statistical significance

The final BERT-GRU + Qwen2.5-0.5B pipeline was tested against both the dense BERT-GRU retriever and the BM25 baseline on the 150 development queries. Because the systems are scored on the same queries the comparison is paired, and because ToT NDCG is far from normal (dominated by 0 and a few 1.0 values) two non-parametric paired tests were used and it requires both to agree: the Wilcoxon signed-rank test [8] and a randomised sign-flip permutation test (10^4 resamples).

On NDCG@1000 the reranker yields a significant gain over the dense bi-encoder ($\Delta=+0.081$; Wilcoxon $p=2.4 \times 10^{-4}$, permutation $p < 10^{-4}$). Of the 50 queries it reorders, 36 improve and 14 worsen, while the remaining 100 are unchanged; the Wilcoxon test is therefore effectively run on those 50 moved queries. Against BM25 the gain is far larger and overwhelmingly significant ($\Delta=+0.173$; Wilcoxon $p=1.0 \times 10^{-9}$, permutation $p < 10^{-4}$).

For R@100 the picture is, by construction, different. The reranker only reorders the dense top-100, so it cannot change which documents fall inside that cut-off: its R@100 is *identical* to the dense retriever’s ($\Delta=0, p=1.0$). This is a correctness check rather than a negative result. The pipeline’s large, significant R@100 advantage over BM25 ($\Delta=+0.247$; Wilcoxon $p=3.7 \times 10^{-7}$, permutation $p < 10^{-4}$) is thus attributable entirely to the dense first stage, not the reranker. Both tests agree on every verdict; the permutation p -values reach the 10^{-4}

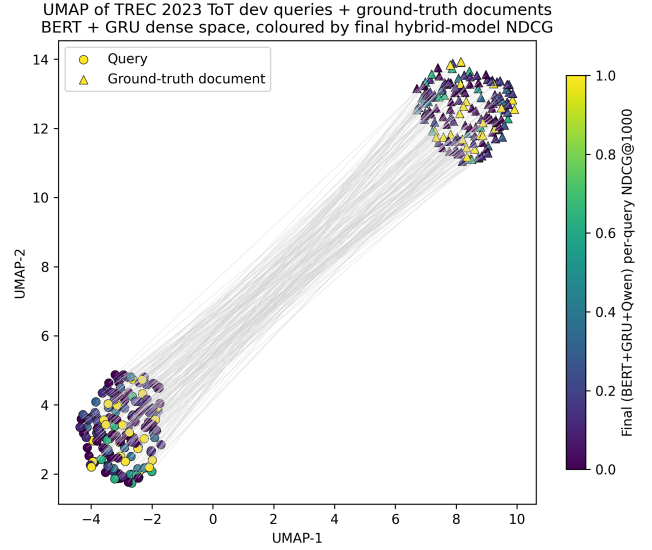


Fig. 3. UMAP of the dense BERT-GRU space: dev queries (circles) and their ground-truth documents (triangles), coloured by final BERT-GRU + Qwen NDCG@1000 (lighter is higher). Queries and documents form two separate regions—the expected dual-encoder modality gap—so the large inter-cluster distance is an artefact of the asymmetric encoders, not a retrieval signal. The interpretable observation is *within* the query cluster: high- and low-NDCG queries are thoroughly interleaved.

resolution floor of 10^4 resamples (i.e. no resampled statistic exceeded the observed one). These results are produced by `significance_test.py` from the cached per-query scores.

VI. INTERPRETABILITY ANALYSIS

To understand *why* the final model retrieves what it does, the 512-dimensional dense embeddings of all 150 dev queries were projected and their ground-truth documents to two dimensions with UMAP ($n_{\text{neighbors}}=30$, $\text{min_dist}=0.1$, cosine metric) [9]. Because the LoRA reranker is a cross-encoder it produces no standalone vector to embed; therefore the dense BERT-GRU space was projected and colours were used to carry the final hybrid-model behaviour. Figure 3 colours each query by its final (BERT-GRU + Qwen) per-query NDCG@1000, and Figure 4 gives a contrastive view that colours each query by the NDCG change the reranker induces.

Two observations follow, both of which are deliberately conservative. First, per-query difficulty is *not* low-dimensionally organised: high- and low-NDCG queries are interleaved within the query cluster (Fig. 3), so retrieval success is driven by specific content match rather than by a query’s coarse position in the embedding space. Second, the reranker’s benefit is likewise diffuse (Fig. 4): it does not rescue an identifiable region but improves a scattered minority of queries, consistent with its role as a top-100 re-orderer. The honest, quantitative summary of its contribution is therefore the per-query delta itself—36 improved, 14 harmed, net $+0.081$ NDCG@1000—rather than any spatial pattern. Because it reorders only within

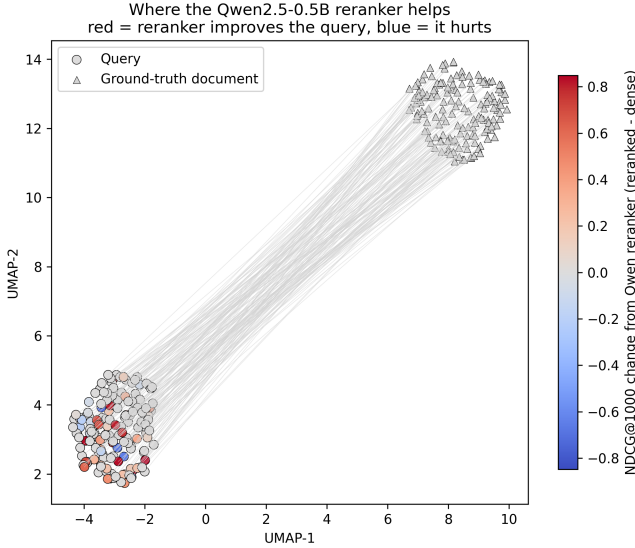


Fig. 4. Contrastive view of the Qwen2.5-0.5B reranker on the same dense space: queries are coloured by their per-query NDCG@1000 change (reranked – dense; red improves, blue hurts, grey unchanged). The reranker improves 36 of 150 queries and harms 14; the effect is diffuse rather than localised to any region of the embedding space.

the dense top-100, this lift comes from promoting relevant documents that were already retrieved, not from recovering queries whose target was never recalled.

VII. ERROR ANALYSIS AND DISCUSSION

To investigate where the model underperforms, the 15 lowest-NDCG@1000 queries were selected under the final pipeline and classified each automatically from the cached per-query scores, using sentence count, a proper-noun heuristic, and the gold document’s position relative to the top-100 rerank pool. The procedure is deterministic and reproducible: no failing query is hand-picked.

The dominant finding is unambiguous: *all 15 of the worst queries are first-stage recall failures*, with NDCG@1000 = 0 and R@100 = 0. The gold document never enters the top-100, so the cross-encoder reranker never scores it. These failures are therefore attributable entirely to the dense retriever, and no reranker—of any size—can recover them; there are *no* ranking errors among the worst queries. This localises the binding constraint of the whole pipeline at the dense first stage.

Crucially, these are *not* failures of corpus coverage or query sparsity. The missed targets are frequently famous, richly documented films—query 403 is *V for Vendetta* (2005), 345 is *A Clockwork Orange* (1971), 1038 is *Smokey and the Bandit II*, 279 is *Freedom Writers* (2007)—all with detailed plot summaries in the collection. Nor does query length help: 403 is an 18-sentence, 16-proper-noun description, among the most detailed in the dev set, yet scores 0 exactly like the one-sentence queries 992 and 902. The limitation is therefore *representational*: the dense bi-encoder cannot bridge the gap between a first-person, scene-by-scene recollection and

TABLE IV
REPRESENTATIVE FAILING QUERY–DOCUMENT PAIRS FROM THE 15 WORST QUERIES (ALL WITH NDCG@1000 = R@100 = 0), GROUPED BY FAILURE FACET.

Query	Gold film	Failure facet
403	<i>V for Vendetta</i>	misremembered attributes
345	<i>A Clockwork Orange</i>	misremembered attributes
832	<i>Mikey</i>	event/scene, no entities
714	<i>Sorority Boys</i>	event/scene, no entities
652	<i>Scary Movie</i>	false/misleading anchor
813	<i>Thriller</i> (TV)	film/TV confusion
861	<i>Nightmares & Dreamscapes</i> (TV)	film/TV confusion

an encyclopedic plot summary, even when the target is well covered and the description is rich.

Reading the isolated query–document pairs exposes four recurring facets of this gap, summarised in Table IV.

- **Misremembered attributes.** Query 403 (*V for Vendetta*, an English/British film) describes it as German or French; 345 (*A Clockwork Orange*) recalls a woman killed during the break-in that the film does not contain. The [CLS]-pooled embedding latches onto these surface attributes—some of them wrong—rather than the film’s underlying plot identity.
- **Event/scene queries without named entities** (832 → *Mikey*, “a kid shoots his father with a bow and arrow”; 714 → *Sorority Boys*). The embedding captures topical gist but not specific plot-event matching against encyclopedic summaries.
- **False or misleading anchors** (652 → *Scary Movie*, “ended the same way as *The Usual Suspects*”). The only strong entity names a *different* film and pulls retrieval toward it; the contrastive objective has no mechanism to discount an asserted-but-wrong attribute.
- **Film/TV content-type confusion**, where the target is a television anthology or series (813 → *Thriller*, 861 → *Nightmares & Dreamscapes*, 625 → *Newhart*) and the query itself hedges that it “may have been a TV show”.

Because every tractable failure is on the recall side and the targets are already well represented in the corpus, the productive next steps address the query–document *representation*, not corpus expansion: a sparse + dense hybrid retriever to recover entity- and title-bearing queries lexically, and retrieval-time query rewriting (e.g. an LLM that normalises a rambling recollection into a canonical plot description and hedges over asserted-but-wrong attributes). Finally, separating gold documents that merely fall outside the top-100 rerank pool (ranks 100–1000, recoverable by widening it) from those missed beyond rank 1000 requires the full ranked run and is the immediate follow-up diagnostic.

VIII. CONCLUSION

This work delivered a full BERT-GRU bi-encoder → Qwen2.5 LoRA reranker pipeline for TREC 2023 Tip-of-the-Tongue retrieval. Across 12 tokenizer configurations, WordPiece-uncased at $d=512$ emerged as the strongest dense

backbone; in the sentence-dropout sweep $p=0.1$ scored highest, although the final pipeline retained the interim $p=0.2$ setting for the reasons noted in Section V-B. The full pipeline reaches NDCG@1000 of 0.2892 and R@100 of 0.4067, with the NDCG@1000 gains over both the BM25 baseline ($p \leq 10^{-9}$) and the dense BERT-GRU retriever ($p < 10^{-3}$) statistically significant under both Wilcoxon and permutation tests; the R@100 advantage over BM25 stems entirely from the dense first stage, since the reranker only reorders within the dense top-100. The error analysis localises the binding constraint at first-stage recall and identifies sparse queries and asserted-but-wrong details as the most tractable next targets, suggesting future work on query rewriting and recall-augmenting hybrid retrieval.

AI USE STATEMENT

AI-assisted grammar checking and LaTeX formatting were used during the preparation of this manuscript. Code-level AI assistance was used while debugging the training loop and the LoRA reranker prompt template. All experimental design, analysis, and interpretation are the author’s own work.

EXPERIMENT TRACKING

All experiments were logged with Weights & Biases, and the following projects are public for reproducibility:

- RQ1 tokenizer/dimension sweep: [tokenizer-sweep-link](#)
- RQ2 sentence-dropout sweep: [tot-rq2-dropout-sweep-link](#)
- LoRA Qwen2.5 reranker training: [tot-reranker-link](#)
- BM25 baseline grid search: [tot-bm25-sweep-link](#)
- Dense BERT-GRU retrieval runs: [tot-retrieval-link](#)

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